Credit Spread Prediction Using Machine Learning

Final Project - MF703

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Introduction

What are Credit Spreads?

- Corporate bond yield minus treasury bond yield
- Measure of market stress and credit risk

Why Does This Matter?

- Guides portfolio allocation
- Helps manage risk and time hedges
- Early signal of economic downturn

Project Objective

- Predict next day U.S. credit spreads using machine learning

What We Built

- Dataset combining macro, market and firm level indicators
- XGBoost and LightGBM model to forecast treasury bond and CCC spreads
- Evaluation and feature importance to understand drivers

Macro Indicators

Credit spreads react to *economic cycles, monetary policy, and market sentiment*

These indicators help capture default risk, liquidity stress, and risk aversion

1. Market Sentiment

- VIX
- S&P 500

2. Economic Conditions

- Unemployment Rate
- GDP Growth
- Industrial Production Growth
- Inflation (CPI)

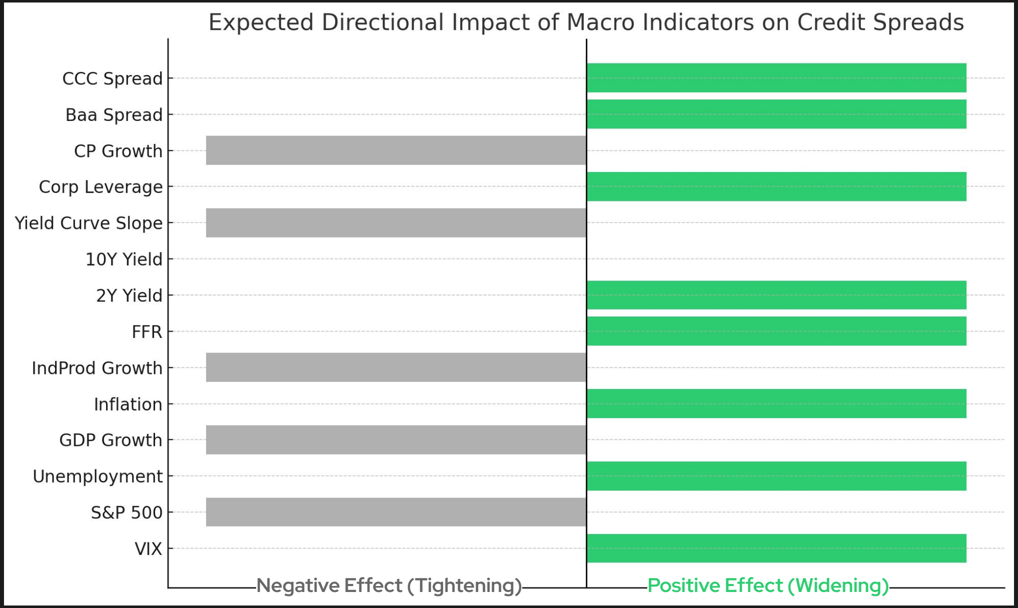
3. Monetary Policy & Rates

- Federal Funds Rate
- 2Y & 10Y Treasury Yields
- Yield Curve Slope (10Y–2Y)

4. Credit Conditions

- Corporate Leverage
- Commercial Paper Growth
- CCC Spread

Macro Indicators



Variables associated with higher uncertainty or weaker economic fundamentals (VIX, unemployment, inflation, and corporate leverage) tend to increase spreads, representing higher perceived default risk.

Conversely, indicators tied to economic strength or market confidence (GDP growth, industrial production, and equity performance) typically tighten spreads

Summary of Variables and Data Sources

Table 1: Credit, Market, and Yield Curve Variables

Variable Name	Description	Source	Frequency
CCC_10Y_Spread	CCC corporate bond yield minus 10Y Treasury yield	FRED	Daily / Monthly
CCC_Yield	Moody's Seasoned CCC Corporate Bond Yield	FRED	Daily
T10Y	10-Year U.S. Treasury yield	FRED	Monthly
T2Y	2-Year U.S. Treasury yield	FRED	Daily
YieldCurveSlope	Term spread (10Y minus 2Y Treasury)	FRED	Daily
VIX	CBOE Volatility Index (close)	Yahoo Finance	Daily
SP500_Return	S&P 500 percentage return	Yahoo Finance	Daily

Table 2: Macroeconomic and Credit Condition Variables

Variable Name	Description	Source	Frequency
Unemployment	U.S. unemployment rate	FRED	Monthly
Inflation	Month-over-month CPI inflation	FRED	Monthly
GDP_Growth	GDP growth rate	FRED	Quarterly
IndPro_Growth	Industrial production growth rate	FRED	Monthly
CP_Growth	Commercial paper outstanding growth	FRED	Quarterly
Corporate_Leverage	Growth in total credit market debt outstanding	FRED	Quarterly

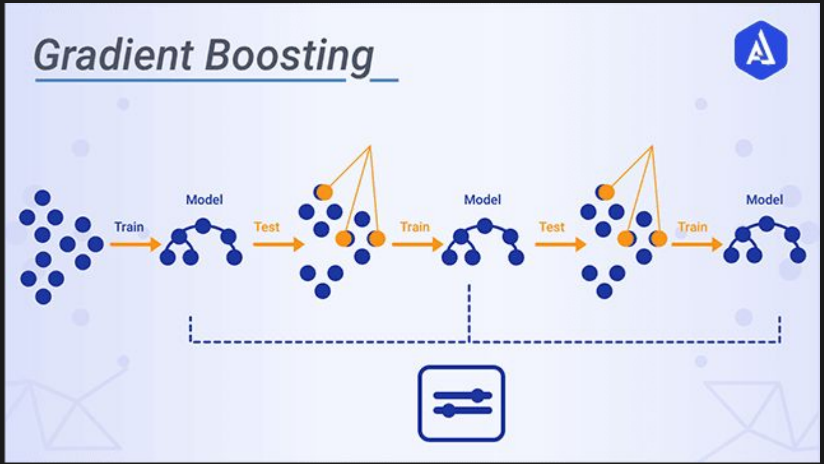
Table provides an overview of the variables included in the empirical analysis. It reports each variable's definition, data source, and original reporting frequency. To enable joint modeling with daily credit spreads, all series are aligned to a daily frequency through interpolation or forward-filling. Feature transformations, including lagged differences and growth rates, are applied to enhance predictive power while avoiding look-ahead bias.

Model Architecture

Gradient Boosting

- Gradient Boosting is an **ensemble method** where models are built sequentially
- Each new tree focuses on **correcting the errors** made by previous trees
- The model improves gradually by minimizing the **residual errors**
- Unlike Random Forest, the trees are **dependent** on one another
- Produces high accuracy on **structured/tabular data**

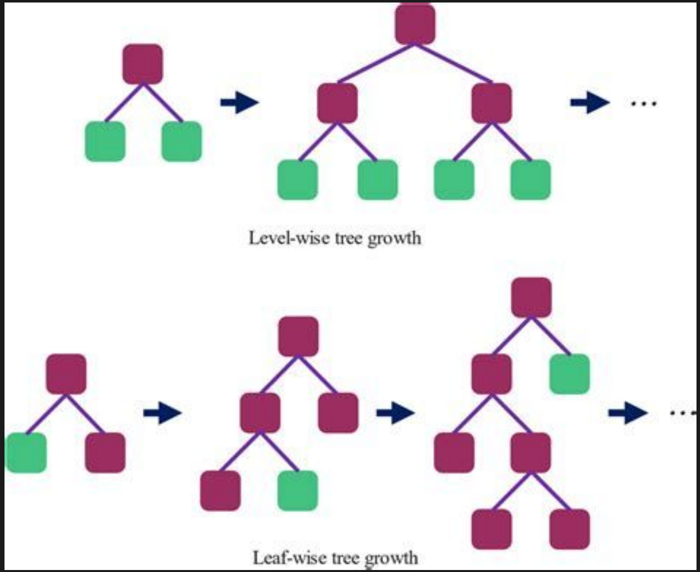
$$\hat{y}_i = F_M(x_i) = \sum_{m=1}^{M} f_m(x_i),$$



XGBoost

$$\mathcal{L}^{(m)} = \sum_{i=1}^{N} L\big(y_i, \hat{y}_i^{(m)}\big) + \sum_{k=1}^{m} \Omega(f_k),$$

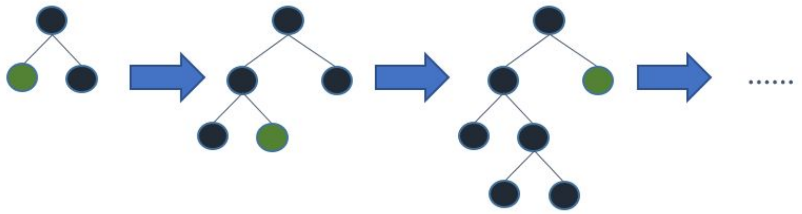
- Uses a **level-wise tree growth** strategy, expanding all nodes at the same depth before moving deeper
- Strong **regularization** (L1 and L2) helps prevent overfitting
- Includes optimized handling of **missing values** and sparse data
- Works extremely well on **small to medium-sized structured datasets**
- Produces stable, balanced trees that generalize reliably



LightGBM

- Uses **leaf-wise tree growth**, always splitting the leaf that produces the largest reduction in error
- This leads to deeper trees in important regions and faster learning of complex patterns
- Employs a **histogram-based algorithm**, grouping feature values to speed up split finding
- Extremely fast and memory-efficient, ideal for **large-scale datasets**
- Supports **categorical features natively** without one-hot encoding
- More prone to overfitting if not tuned carefully due to its aggressive splitting strategy

$$\Delta\mathcal{L} = \frac{1}{2}\left(\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right) - \gamma,$$



Leaf-wise tree growth

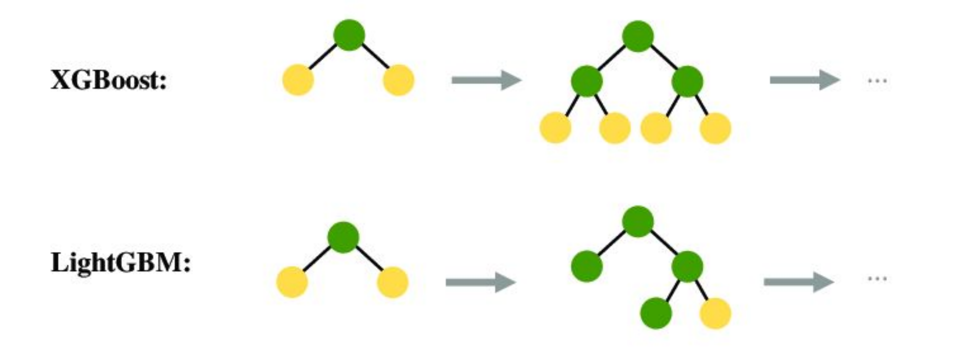
Pros and Cons

Pros & Cons — XGBoost

- **Pros:** Stable, reliable, strong regularization, great performance on smaller datasets
- **Cons:** Slower on very large datasets, higher memory usage

Pros & Cons — LightGBM

- **Pros:** Extremely fast, handles large datasets well, efficient memory usage, native categorical support
- **Cons:** Can overfit more easily, requires tuning, tree structure less controlled



Data Handling & Feature Engineering

1. Data Preparation

- Daily alignment of all macro, market, and credit series
- Sourced from FredApi and yfinance date-time normalized
- Missing values handled via forward-fill and interpolation
- No look-ahead bias: features use information available at time t

2. Synthetic Features Created

- Lagged variables (e.g., VIX_{t-1}, GDP_{t-1}) to capture delayed macro effects
- Yield Curve Slope = 10Y − 2Y (recession indicator)
- Spread Differentials (CCC - Treasury bond yields)
- Growth rates / percentage changes for activity indicators
- Volatility proxies from market data

3. Justification

- Improves temporal predictability in credit spread movements
- Captures nonlinear relationships ML models rely on
- Enhances the model's ability to detect early signs of economic stress
- Reduces noise and structural breaks by working with transformed data

Hyper parameters

LightGBM

Key Hyperparameters

- **n_estimators:** 3,000
 Sufficient boosting rounds to refine predictions with a moderate learning rate.
- **learning_rate:** 0.02
 Enables stable, incremental learning.
- **max_depth:** -1
 Unlimited depth; tree size effectively governed by `num_leaves`.

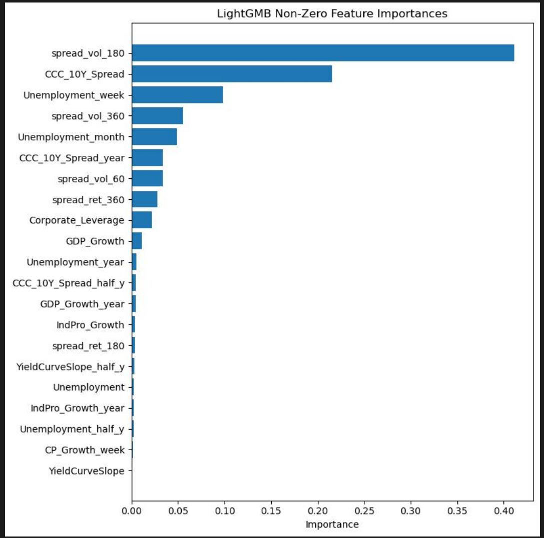
XGBoost

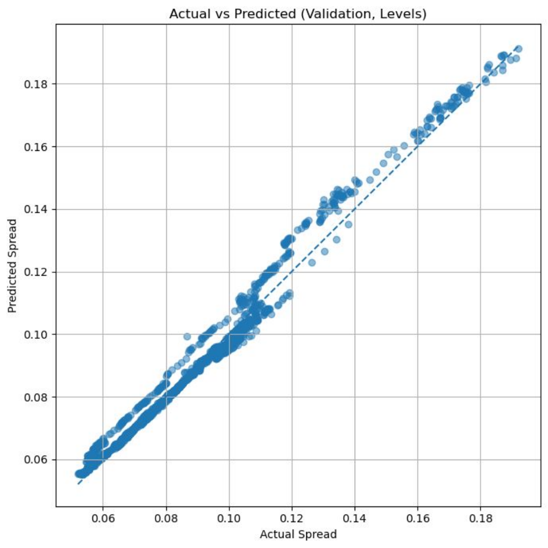
Key Hyperparameters

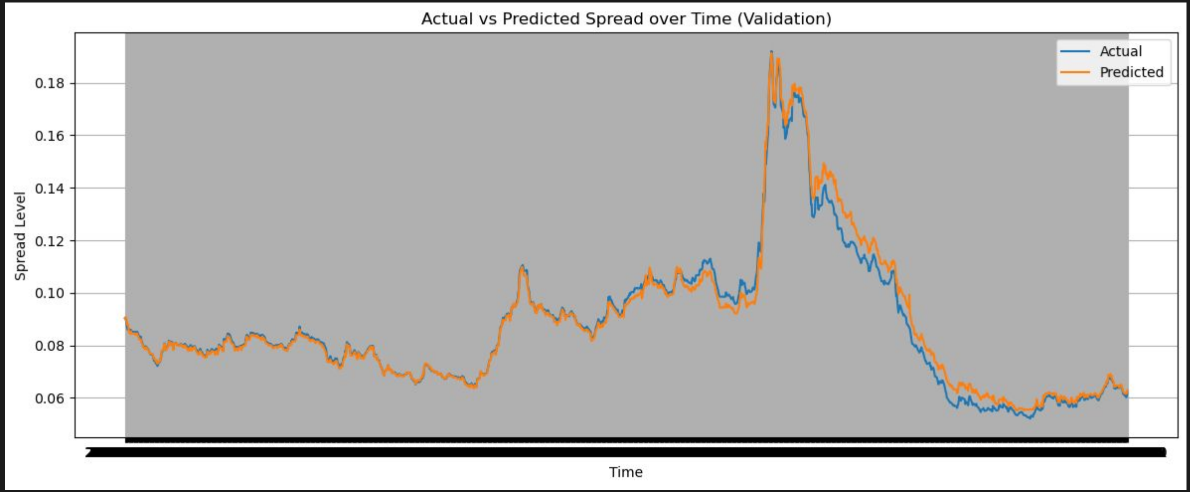
- **n_estimators:** 3,000
 Large number of boosting rounds to improve predictive accuracy with a low learning rate.
- **learning_rate:** 0.01
 Enables gradual learning and reduces risk of overfitting.

Model Results

LightGBM

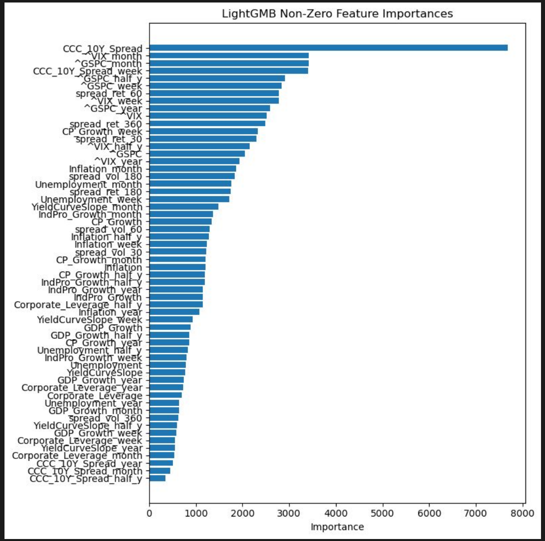


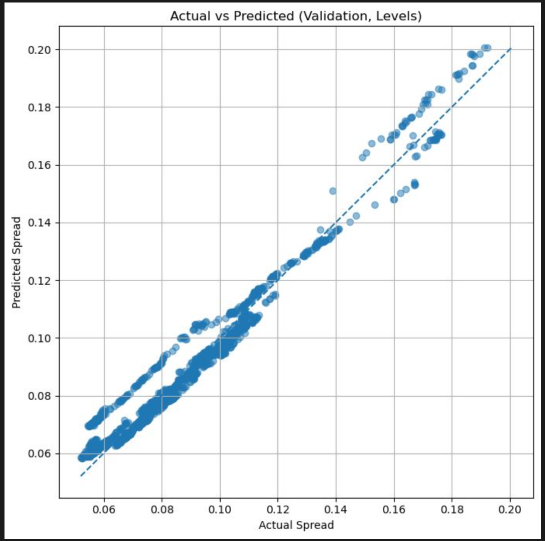


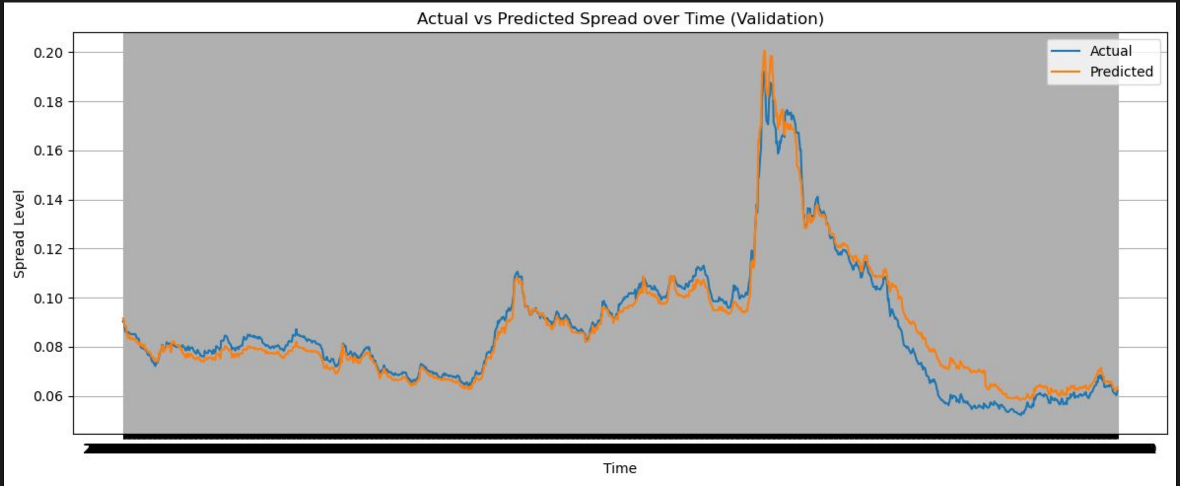


Model Results

XGBoost







Practical Applications

Early warning

- Allows investors to de-risk by signalling when credit spread will rise

Portfolio Allocation

- Guides how to allocate treasury bonds vs risky bonds

Timing Hedges

- Hedging or changing credit exposure depending on predicted spread level

Stress Testing

- Asset managers can use model to test scenarios

Corporate Financing

- Helps firms decide when to issue new bonds or refinance debt

Policy Insight

- Policymakers can understand how economic shifts transmit into credit spread changes

Future Works

Regime Switching

- Crisis vs Normal mode can increase accuracy during shocks

Global Macro Variables

- Global PMI, oil prices, geopolitical risk index

Multi-Horizon Forecasting

- Predict longer horizons such as weekly, monthly, semi-annual

Time Series Predictions

- Predict price fluctuations for a given time period

Thank you

Q&A